MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

04

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. 1

Choice A is correct. Applying the distributive property on the left- and right-hand sides of the given equation yields $x^2+2x+3x+6=x^2-2x-3x+6+10$, or $x^2+5x+6=x^2-5x+16$. Subtracting x^2 from and adding $5x$ to both sides of this equation yields $10x+6=16$. Subtracting 6 from both sides of this equation and then dividing both sides by 10 yields $x=1$.

Choices B, C, and D are incorrect. Substituting 0, -2, or -5 for x in the given equation will result in a false statement. If $x=0$, the given equation becomes $6=16$; if $x=-2$, the given equation becomes $0=30$; and if $x=-5$, the given equation becomes $6=66$. Therefore, the values 0, -2, and -5 aren't solutions to the given equation.

Q2

A

A. 1

Choice A is correct. Substituting x^2 for y in the second equation gives $2(x^2)+6=2(x+3)$. This equation can be solved as follows:

$2x^2+6=2x+6$	Apply the distributive property.
$2x^2+6-2x-6=0$	Subtract 2x and 6 from both sides of the equation.
$2x^2-2x=0$	Combine like terms.
$2x(x-1)=0$	Factor both terms on the left side of the equation by 2x.

Thus, $x=0$ and $x=1$ are the solutions to the system. Since $x>0$, only $x=1$ needs to be considered. The value of y when $x=1$ is $y=x^2=1^2=1$. Therefore, the value of xy is $(1)(1)=1$.

Choices B, C, and D are incorrect and likely result from a computational or conceptual error when solving this system of equations.

Q3

3

The correct answer is 3. The solution to the given equation can be found by factoring the quadratic expression. The factors can be determined by finding two numbers with a sum of 1 and a product of -12. The two numbers that meet these constraints are 4 and -3. Therefore, the given equation can be rewritten as $(x+4)(x-3)=0$. It follows that the solutions to the equation are $x=-4$ or $x=3$. Since it is given that $a>0$, a must equal 3.

Q4

C

C. $2, 0$

Choice C is correct. The second equation in the given system of equations is $y = 6x - 12$. Substituting $6x - 12$ for y in the first equation of the given system yields $6x - 12 = x - 2x + 4$. Factoring 6 out of the left-hand side of this equation yields $6x - 2 = x - 2x + 4$. An expression with a factor of the form $x - a$ is equal to zero when $x = a$. Each side of this equation has a factor of $x - 2$, so each side of the equation is equal to zero when $x = 2$. Substituting 2 for x into the equation $6x - 2 = x - 2x + 4$ yields $62 - 2 = 2 - 22 + 4$, or $0 = 0$, which is true. Substituting 2 for x into the second equation in the given system of equations yields $y = 62 - 12$, or $y = 0$. Therefore, the solution to the system of equations is the ordered pair $2, 0$.

Choice A is incorrect and may result from switching the order of the solutions for x and y.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

C

C. $x = \frac{y}{77b}$

Choice C is correct. Multiplying each side of the given equation by y yields the equivalent equation $\frac{y}{7b} = 11x$. Dividing each side of this equation by 11 yields $\frac{y}{77b} = x$, or $x = \frac{y}{77b}$.

Choice A is incorrect. This equation is not equivalent to the given equation.

Choice B is incorrect. This equation is not equivalent to the given equation.

Choice D is incorrect. This equation is not equivalent to the given equation.

Q6

D

D. -8

Choice D is correct. By the zero-product property, if $(x-4)(x+2)(x-1)=0$, then $x-4=0$, $x+2=0$, or $x-1=0$. Solving each of these equations for x yields $x=4$, $x=-2$, or $x=1$. The product of these solutions is $(4)(-2)(1)=-8$.

Choice A is incorrect and may result from sign errors made when solving the given equation. Choice B is incorrect and may result from finding the sum, not the product, of the solutions. Choice C is incorrect and may result from finding the sum, not the product, of the solutions in addition to making sign errors when solving the given equation.

Q7

A

A. $\dfrac{-5 \pm \sqrt{25+168}}{12}$

Choice A is correct. The quadratic formula, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, can be used to find the solutions to an equation in the form $ax^2 + bx + c = 0$. In the given equation, $a = 6$, $b = 5$, and $c = -7$. Substituting these values into the quadratic formula gives $\dfrac{-5 \pm \sqrt{5^2 - 4(6)(-7)}}{2(6)}$, or $\dfrac{-5 \pm \sqrt{25+168}}{12}$.

Choice B is incorrect and may result from using $\dfrac{-a \pm \sqrt{b^2 - 4ac}}{2a}$ as the quadratic formula.

Choice C is incorrect and may result from using $\dfrac{-b \pm \sqrt{a^2 + 4ac}}{2a}$ as the quadratic formula.

Choice D is incorrect and may result from using $\dfrac{-a \pm \sqrt{a^2 + 4ac}}{2a}$ as the quadratic formula.

Q8

A

A. $4j+9=\frac{k}{p}$

Choice A is correct. To express $4j+9$ in terms of p and k, the given equation must be solved for $4j+9$. Since it's given that j is a positive number, $4j+9$ is not equal to zero. Therefore, multiplying both sides of the given equation by $4j+9$ yields the equivalent equation $p4j+9=k$. Since it's given that p is a positive number, p is not equal to zero. Therefore, dividing each side of the equation $p4j+9=k$ by p yields the equivalent equation $4j+9=\frac{k}{p}$.

Choice B is incorrect. This equation is equivalent to $p=\frac{4j+9}{k}$.

Choice C is incorrect. This equation is equivalent to $p=k-4j-9$.

Choice D is incorrect. This equation is equivalent to $p=k4j+9$.

Q9

D

D. $12t$

Choice D is correct. Subtracting b from each side of the given equation yields $12t=c-b$. Therefore, the expression $12t$ correctly expresses the value of $c-b$ in terms of t.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10 D

D. $-6 + 2\sqrt{19}$

Choice D is correct. Adding 40 to both sides of the given equation yields $w^2 + 12w = 40$. To complete the square, adding $\frac{12}{2}^2$, or 6^2, to both sides of this equation yields $w^2 + 12w + 6^2 = 40 + 6^2$, or $w + 6^2 = 76$. Taking the square root of both sides of this equation yields $w + 6 = \pm\sqrt{76}$, or $w + 6 = \pm 2\sqrt{19}$. Subtracting 6 from both sides of this equation yields $w = -6 \pm 2\sqrt{19}$. Therefore, the solutions to the given equation are $-6 + 2\sqrt{19}$ and $-6 - 2\sqrt{19}$. Of these two solutions, only $-6 + 2\sqrt{19}$ is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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